UGANDA ADVANCED CERTIFICATE OF EDUCATION

AGRICULTURE SET FOUR ASSESSMENT 2026

SENIOR SIX

Paper 1 (theory)

3 Hours

INSTRUCTIONS TO CANDIDATES:

This paper consists of **6** *examination items. It has* **two** *sections;* **A** *and* **B**.
Section **A** *has two compulsory items.*
Section **B** *has* **two** *parts;* **I** *and* **II**. *Answer* **one** *item from each part.*
Answer four examination items in all.
Indicate items attempted in ***Section B (Part I*** *&* ***Part II)***
All *answers* **must** *be written in the Answer sheets provided.*

ITEM.			**FOR EXAMINERS USE ONLY (SCORES).**
1			
2			
SECTION B	**Part I**		
	Part II		
TOTAL SCORES			

SECTION A.

Attempt all items in this section.

Item 1

Nabirye is a fish farmer near Lake Kyoga where she rears tilapia and catfish followed by selling them when they are fresh. During periods of high harvest, she experiences heavy losses because fresh fish spoils before reaching distant markets. Fish prices also fall whenever many farmers harvest at the same time. Most customers in nearby towns prefer ready-to-cook fish products

Recently, a supermarket chain, several schools and hotels expressed interest in buying fish sausages if she could supply them consistently. Nabirye would like to take advantage of this opportunity to increase her income.

She sells fish individually without working with other fish farmers. The fish are graded according to weight by visual examination for better pricing. She packs her fish in clear and clean used food containers. Most of her sales are made through middlemen who connect him to other consumers. She transports fish in clean basins to nearby towns.

Unsold fish is stored near the fire place in the kitchen. She advertises her products using word of mouth throughout the entire area for simplicity. She relies on boda-boda men for market information and prevailing prices in different markets. As a result, her fish products attract few customers and her profits remain low.

Task

(a) Guide Nabirye on how she can process fish sausages for sale to the supermarkets, schools and hotels.
(b) Explain the measures Nabirye should take to improve the marketing and profitability of her fish products.

Item 2

Ninsiima operates a dairy cattle enterprise and a maize farm in Mbarara District. To increase milk production, she formulated her own feeds using maize grain and other locally available ingredients. Before feeding the cattle, she boiled the maize mixture for a long time and mixed it with a commercial enzyme supplement while it was still very hot. However, the animals continued to show poor feed utilization and passed large amounts of undigested material in their dung.

During a prolonged dry season, Ninsiima supplied her cattle with highly concentrated mineral solutions. Some animals later showed signs of dehydration despite regular feeding.
To improve her herd, Ninsiima selected a breeding bull mainly because it had a beautiful coat colour. Some of the daughters produced by the bull also showed low productivity. In another breeding programme, she repeatedly mated closely related animals for several generations. Over time, fertility declined and calf mortality increased.

One heifer failed to conceive despite repeated mating. A veterinary officer later discovered abnormalities in her reproductive system. Other cows were often served several days after showing heat signs.
Several calves showed poor growth despite receiving adequate feed. On examination, the animals had poorly developed rumens because they had been maintained mainly on milk for a long period. In addition, some animals bled excessively after sustaining minor injuries and the bleeding took a long time to stop.

Task

a) Explain the causes of low returns from Ninsiima's enterprises.
b) Suggest, with reasons, how Ninsiima can improve returns from his enterprises.

SECTION B.
Part I. ANIMAL PRODUCTION
Choose one item from this section.

Item 3

After identifying a growing demand for goat meat in the surrounding local markets, Mr. Kamau established a goat enterprise on his 5-acre open farmland. He purchased goats from different sources, and some were noticeably smaller and weaker than others at the time of stocking. Within the housing unit, some goats stood for long periods and in corners, while others had soiled side coats.

A strong odour was noticeable whenever one entered the shed. Males and females grazed together throughout the year. At the end of each year, the farmer was unable to clearly account for the income and expenditure of the enterprise. During grazing, some goats spent long hours feeding, while others moved around the farm in search of more palatable grasses. Some goats were observed with bleeding hooves. Several animals developed rough hair coats, and distended abdomens.

On certain occasions, intramuscular drugs were administered, and two days later, some mature goats were slaughtered in a banana plantation. During slaughter, some carcasses showed bruises and fractured bones while workers were observed with bear hands during slaughtering.

Task.

a) Explain the causes of the observations on the farm and the effect of the farm practices at the farm on the performance of fish.
b) With reasons, suggest the measures that should be taken to improve production on the farm.

Item 4

After receiving Youth Livelihood Funds, a youth group in Kairo District decided to invest in fish farming as an income generating activity. They constructed one large-sized pond near a seasonal swamp called Naigombwa.

The pond was dug at varying depths. Occasionally, animals were seen entering the pond. During the rainy season, the pond flooded, and water stagnated in it for about three days after a heavy downpour. Three months after stocking, the pond depth reduced, and it became noticeably shallower. The group stocked the pond with both catfish and tilapia fingerlings. One month after stocking, many fish were seen swimming near the pond surface, and some were observed scraping against the pond walls.

Four months after stocking, the pond water became dark and cloudy, and algae accumulated on the surface. Some rotting fish feed was seen accumulating at the bottom of the pond. At harvest, a well sized mosquito net was used for harvesting during a hot afternoon, and some fish were smaller than the expected market size.

Task.

c) Explain the causes of the observations on the farm and the effect of the farm practices at the farm on the performance of fish.
d) With reasons, suggest the measures that should be taken to improve production on the farm.

Part II, CROP PRODUCTION.

Choose one item

Item 5

Mr. Okello acquired 12 hectares of land under customary arrangements in a semi-arid region. The land had previously been used for uncontrolled grazing. He cleared the vegetation by burning and ploughed the soil once before planting cassava cuttings obtained from farmers in areas that receive high annual rainfall. Two months after planting, foxtail grew taller than the cassava plants between rows. Crop residues were removed after harvest, and fertilizer was applied occasionally without prior soil testing. The crop relied entirely on rainfall.

During the wet season, water sometimes gets stranded in the garden briefly after rainfall. The cassava plants developed curled leaves, smaller stems with nearly fused internodes and reduced tuber size. At harvest, tubers were harvested by digging them out using a hoe in a dry surface when the leaves were entirely green. Harvested tubers are dropped on the surface after harvesting.

Task

a) Explain the causes of the observations on Mr. Okello's farm and the effect of the farm practices at his farm on the yield of the cassava.
b) With reasons, suggest the measures that should be taken to improve production on Mr. Okello's farm.

Item 6.

After receiving PDM funds, Mr. Nsubuga established a pineapple plantation on 2 acres of sloping land, one section of which had marshy vegetation, while the other dried quickly and developed surface cracks during dry periods. The land was cleared using surface tillage before planting pineapple suckers as obtained from neighbouring farms.

Three months after planting, some plants appeared closely packed, while a mixture of maize, sorghum, and bean plants grew vigorously in the garden. In some parts of the field, the soil surface was bare and exposed, whereas in others it was covered with a thick layer of dry plant materials. Some leaned towards sunlight.
On several plants, leaves showed small holes and scotched margins, while others have yellow strips between the leaf margins. Towards harvesting, maturity of fruits is mainly judged by estimating the date of planting. During harvesting, some fruits had split eyes and a strong fermented smell, while others were observed with deep knife cuts and torn stalk attachments after harvesting.

Task;

a) Explain the causes of the observations on Mr. Nsubuga's farm and the effect of the farm practices at his farm on the yield of the pineapples.
b) With reasons, suggest the measures that should be taken to improve production on Mr. Nsubuga's farm.

END